SGMNet: Scene Graph Matching Network for Few-Shot Remote Sensing Scene Classification

Baoquan Zhang, Shanshan Feng, Xutao Li, Yunming Ye, Rui Ye, Chen Luo, and Hao Jiang

***Abstract*— Few-shot remote sensing scene classification (FSRSSC) is an important task, which aims to recognize novel scene classes with few examples. Recently, several studies attempt to address the FSRSSC problem by following few-shot natural image classification methods. These existing methods have made promising progress and achieved superior performance. However, they all overlook two unique characteristics of remote sensing images: 1) *object cooccurrence* that multiple objects tend to appear together in a scene image and 2) *object spatial correlation* that these cooccurrence objects are distributed in the scene image following some spatial structure patterns. Such unique characteristics are very beneficial for FSRSSC, which can effectively alleviate the scarcity issue of labeled remote sensing images since they can provide more refined descriptions for each scene class. To fully exploit these characteristics, we propose a novel scene-graph-matching-based meta-learning framework for FSRSSC, called SGMNet. In this framework, a scene graph construction module is carefully designed to represent each test remote sensing image or each scene class as a scene graph, where the nodes reflect these cooccurrence objects; meanwhile, the edges capture the spatial correlations between these cooccurrence objects. Then, a scene graph matching module is further developed to evaluate the similarity score between each test remote sensing image and each scene class. Finally, based on the similarity scores, we perform the scene class prediction via a nearest neighbor classifier. We conduct extensive experiments on the UCMercedLandUse, WHU19, AID, and NWPU-RESISC45 datasets. The experimental results show that our method obtains superior performance over the previous state-of-the-art methods.**

***Index Terms*— Deep neural networks, few-shot image classification, metric learning, remote sensing, scene classification.**

I. Introduction

REMOTE sensing scene classification (RSSC) is an important research problem on remote sensing applications, such as disaster detection [1], urban planning [2], and land use [3]. It aims to recognize a remote sensing image to a scene semantic class. During the past years, RSSC has made rapid advances such as [4], [5], relying on convolutional neural network (CNN) and a large number of labeled data. However, unlike the annotation task of natural images [6], the annotation task of remote sensing images is very time-consuming and laborious since it needs to be achieved by experts with rich knowledge of remote sensing domains [7]. To alleviate the burden of annotating data, the problem of few-shot remote sensing scene classification (FSRSSC) has been proposed recently, which draws wide attention. Different from RSSC, FSRSSC aims to learn task-agnostic meta-knowledge from base scene classes with abundant labeled samples and then transfer the meta-knowledge to recognize novel scene classes with limited labeled samples [7], [8], [9], where these novel scene classes are not overlapped with the base scene classes.

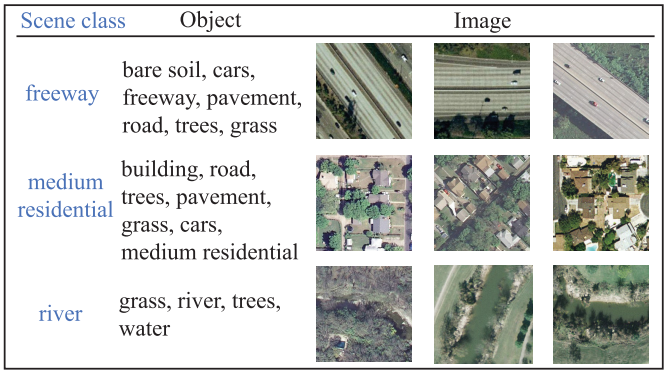


Fig. 1. Unique characteristics of remote sensing images. The first one is object cooccurrence, e.g., "cars" and "road" tend to appear together in the scene image of "freeway," and "grass" and "water" jointly occur in the scene class of "river." The second one is object spatial correlation, e.g., "cars" are usually running or parked on the "road." These characteristics can provide more refined descriptions for remote sensing scene images/classes.

Few-shot natural image classification (FSNIC) problem is closely related to FSRSSC, which aims to quickly recognize novel natural classes from very few examples [10], [11], [12], [13]. The main difference is that the former focuses on natural images, whereas the latter targets at remote sensing scene images. At present, a large number of FSNIC methods have been proposed from the perspective of meta-learning [11], [14], [15]. These methods can be roughly grouped into two categories: 1) metric-based methods that perform novel class prediction via a nearest neighbor classifier with the Euclidean [16], cosine [17], or learnable distance [12] and 2) optimization-based methods that learn a new optimization algorithm over few labeled samples to quickly train a classifier for novel classes [18], [19], [20]. Although these methods have shown superior performance on FSNIC, all the methods

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focus on recognizing natural images instead of remote sensing images. Compared with natural images, remote sensing images are more challenging to classify since they have larger intra-class variances and smaller interclass differences [7], [8], [21], [22], [23]. Thus, the classification performance of these FSNIC methods is unsatisfactory when they are directly applied to FSRSSC.

Recently, several studies [8], [21] attempt to extend these existing FSNIC methods to recognize remote sensing scene images. Their basic idea is designing a good feature extractor [8], [21], metric mechanism [7], and hyperparameter (e.g., the learning rate) of the optimization algorithm [24] for FSRSSC. Although these FSRSSC methods have achieved promising improvements, they all overlook two intrinsic characteristics of remote sensing images that are different from natural images: 1) **object cooccurrence** refers to that the remote sensing scene images usually contain multiple objects and they usually appear together in a scene image. The cooccurrence objects are usually different in different scene classes. For example, as shown in Fig. 1, the objects "road" and "car" tend to occur together in the scene image of "freeway," and "grass" and "water" often accompany the scene class of "river." In fact, object cooccurrence has been investigated in the domain of traditional remote sensing image classification [25], [26] where a large number of labeled samples are available, but as far as we know, there is no previous work that has explored it on FSRSSC and 2) **object spatial correlation** refers to that these cooccurrence objects are distributed in the scene image following some spatial structure patterns, which means that these objects may be correlated with each other. The spatial patterns of objects in different scene classes are also different. For instance, as shown in Fig. 1, the medium residential scene can be described with the spatial structure of cooccurrence objects: 1) the "tree" usually lines both sides of the "road"; 2) the "car" is running or parked on the "road"; 3) the "building" is usually surrounded by the "roads"; and 4) a large amount of "grass" is distributed near the "building." Such characteristics are very beneficial for FSRSSC, which can effectively alleviate the data scarcity issue because they can provide more refined descriptions for each scene class.

To fully leverage these two characteristics (i.e., object cooccurrence and object spatial correlation) of remote sensing images for FSRSSC, in this article, we propose a scene-graph-matching-based meta-learning framework, called SGMNet. Our key idea is encoding each test remote sensing image or each scene class as a scene graph (instead of a simple feature vector) and then evaluating their similarity in a scene graph matching manner. The advantage of such scene-graph-based design is that it can provide more spatial structural information for evaluating the similarity between each test remote sensing image and each scene class. This is because the scene graph nodes can effectively reflect these cooccurrence objects; meanwhile, the spatial correlation of these objects can be explicitly captured by the scene graph edges, such that each remote sensing image/class can be more finely characterized.

To this end, we first pretrain a fully convolutional network on the base scene classes to obtain a good spatial representation for each remote sensing image. Second, given an FSRSSC task, we obtain the spatial representation of each scene class by averaging the spatial representation of all the labeled samples from the class. Then, a scene graph matching network is designed to evaluate the similarity score between each test image and each scene class, which consists of a graph construction module (GCM) and a graph matching module (GMM). Here, the former is in charge of mapping the spatial representation of each test image or each scene class to a scene graph by encoding its local features as nodes and the relationship between all the local features as edges. The latter accounts for evaluating their similarity score at the scene graph level. Finally, based on the similarity scores, we perform the prediction of novel scene classes using a nearest neighbor classifier for FSRSSC.

Our main contributions can be summarized as follows.

1) For FSRSSC, we identify two significant remote sensing image characteristics, ignored in the existing FSRSSC methods: object cooccurrence and object spatial correlation. Resorting to these characteristics, we can effectively alleviate the data scarcity issue. To our knowledge, this is the first work to explore the two characteristics for FSRSSC.
2) We propose a novel scene-graph-matching-based meta-learning framework for FSRSSC. In the framework, a GCM and a graph match module are designed, which fully use the two characteristics of object cooccurrence and object spatial correlation and offer our framework the excellent ability to perform similarity evaluation of remote sensing scenes.
3) We conduct comprehensive experiments on four real-world remote sensing datasets. The experimental results demonstrate that the proposed method achieves superior performance over various state-of-the-art approaches.

The remaining of this article is organized as follows: In Section II, we discuss some related works on RSSC, FSNIC, FSRSSC, and graph matching techniques. In Section III, we describe the proposed SGMNet framework in detail, including an overall framework, a GCM, and a GMM. Section IV shows and analyzes the experimental results on four real-world RSSC datasets. Finally, the conclusion is presented in Section V.

II. Related Work

This article is related to four research domains, including RSSC, FSNIC, FSRSSC, and graph matching techniques, respectively. Next, we review them in detail, respectively.

A. Remote Sensing Scene Classification

RSSC aims to recognize each remote sensing image with a scene semantic class, which has gained wide attention in the past few decades [29], [30]. In the early stage, most existing methods rely on human engineering features (e.g., scale-invariant feature transformation (SIFT) [31], texture descriptors (TDs) [32], and bag-of-visual-words (BOVW) [33]) and the traditional machine learning methods (e.g., support vector machine (SVM) [34]) to perform scene prediction for each

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remote sensing image. For example, in [35], Zhu *et al.* proposed to combine the SIFT and TD features and then used an SVM classifier to predict the scene classes. Zhao *et al.* [36] proposed a rotation-invariant representation model, incorporated it into the BOVW method, and then performed scene prediction by SVM.

Although these methods have made some progress in RSSC, their performance improvement depends heavily on the quality of human engineering features, which limits their performance improvement. To address the drawback, a type of novel RSSC approaches based on deep learning is proposed. This type of methods can automatically learn discriminative features from large amounts of labeled data in an end-to-end train manner. For instance, in [37], Penatti *et al.* first introduced a CNN to address the RSSC problem and achieved promising classification performance. Cheng *et al.* [38] proposed a novel bag of convolutional features (called BoCF) for RSSC by leveraging the pretrained CNN features to replace these traditional descriptors. Wang *et al.* [39] explored the attention mechanism and designed a novel attention-based recurrent convolutional network to address the RSSC problem. Wang *et al.* [30] proposed a global–local two-stream architecture to learn multiscale representation for RSSC. Xu *et al.* [40] developed a graph-convolution-based deep feature aggregation framework to produce more refined features for RSSC.

Different from these RSSC studies, our work 1) focuses on the FSRSSC problem where only few labeled samples are available for each scene class and 2) proposes a novel scene-graph-based metric strategy to evaluate the similarity score between each remote sensing image and each scene class, which fully exploits object cooccurrence and spatial correlation of remote sensing images for FSRSSC.

B. Few-Shot Natural Image Classification

FSNIC aims to learn transferable meta-knowledge from base natural classes with sufficient labeled samples and then leverage it to recognize novel natural classes with few examples. Recently, a large number of FSNIC methods have been proposed, which can be roughly divided into two groups.

1) Metric-based methods. The idea behind this type of method is learning a good metric space where the few-shot classification task can be addressed by a nearest neighbor classifier with the Euclidean distance [16], cosine distance [17], [41], Earth mover's distance [42], and learnable distance [12]. For example, Snell *et al.* [16] averaged the features of all the labeled samples to represent each class prototype and then assigned each test sample to the labels of its nearest class prototype.

2) Optimization-based methods. This line of methods attempts to model an optimization algorithm over few labeled samples under the meta-learning framework [20], [43], [44], aiming to adapt to novel natural classes by a few gradient updates. This goal is achieved by meta-learning the hyperparameters of the conventional optimization algorithm such as model initialization [14], update rule [45], learning rate [46], or weight decay [18]. For example, Finn *et al.* [14] proposed a two-loop optimization framework where the inner loop is in charge of fine-tuning an initial classifier to recognize novel natural classes, and the outer loop accounts for improving the generalization ability of the initial classifier.

Different from these existing methods, our work 1) focuses on recognizing remote sensing images instead of natural images, which have larger intraclass variances and smaller interclass differences, and 2) proposes a novel scene-graph-matching-based meta-learning framework for FSRSSC.

C. Few-Shot Remote Sensing Scene Classification

FSRSSC is closely related to FSNIC, which aims to recognize novel scene classes with few examples. Recent works attempt to address the FSRSSC problem by following the idea of FSNIC. Similarly, these methods can also be roughly divided into two groups.

1) Metric-based methods. The group of works extended the metric-based meta-learning method to address the FSRSSC problem [7], [8], [9], [21], [22], [23]. For example, in [8], Alajaji *et al.* extended the prototypical network [16] by introducing a pretraining strategy on all the base scene classes and performing novel scene class prediction by a cosine nearest-centroid classifier. Li *et al.* [21] incorporated the channel attention and spatial attention modules on the feature network and proposed the feature fusion schemes to achieve discriminative features for FSRSSC. In [7], a new learnable metric module and a novel loss function were developed for FSRSSC.

2) Optimization-based methods. In [47], the authors extended the MAML method proposed in [14] to address the FSRSSC problem and verified its effectiveness. Besides, in [48], Rußwurm *et al.* also extend MAML, but they focus on addressing the cross-cities' few-shot land cover classification problem. Their goal is learning meta-knowledge from many cities with abundant data and then transferring to many cities lacking data to recognize the land cover.

Different from these methods, our work explores the characteristics of remote sensing images, i.e., object cooccurrence and object spatial correlation, by viewing each image or each scene class as a scene graph, and focuses on learning a better similarity evaluation for nearest-class classifier in a scene-graph-matching-based manner.

D. Graph Matching

Graph matching aims to evaluate the similarity between two graphs by encoding their structural information including graph node features and complex node relationships. Typically, graph matching can be formulated as a quadratic assignment problem [49], [50], i.e., assigning each node in the first graph to a unique node in the second graph [51], [52]. Traditional graph matching, such as random walk [53] and spectral matching [54], mainly addresses the problem from the view of optimization heuristics [55], [56]. Although these methods have shown promising performance, all these optimization methods rely on handcrafted features [57], which limits their performance. Recently, deep-learning-based graph matching methods have been proposed and they showed superior performance on graph matching. For example, in [58],

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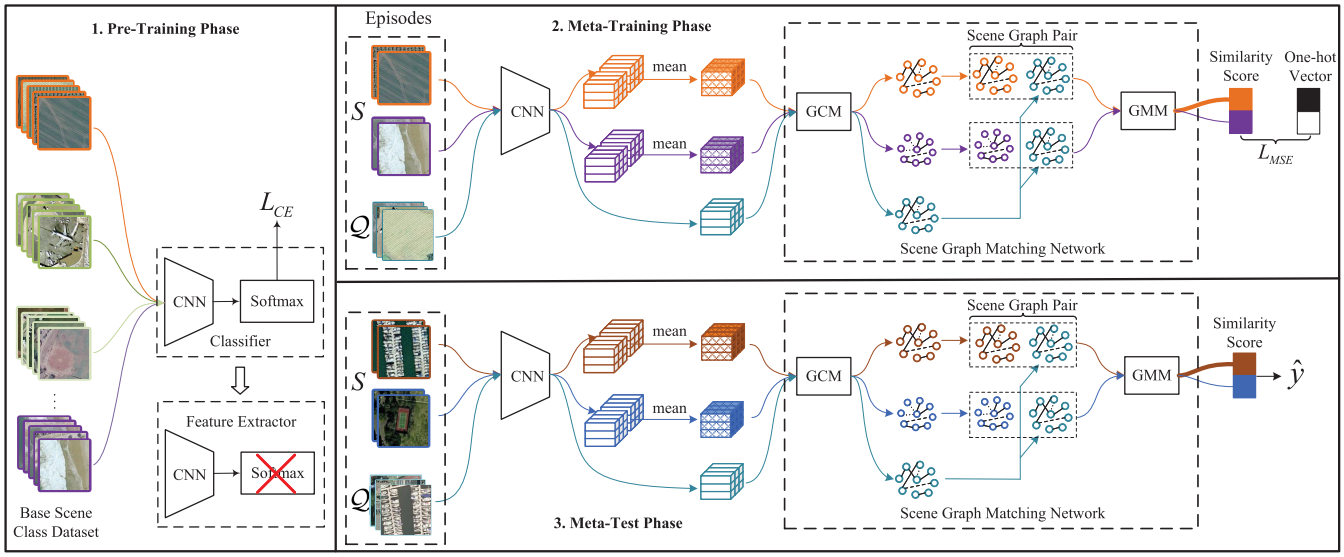


Fig. 2. Scene-graph-matching-based meta-learning framework for FSRSSC, including three phases. (1) Pretraining phase that learns a fully convolutional-network-based feature extractor on the base scene classes (Section III-B1). (2) Meta-training phase that learns a scene-graph-level metric strategy in an episodic training manner (Section III-B3). (3) Meta-test phase that performs novel class prediction by constructing and matching scene graphs (Section III-B3).

a data-driven graph matching network was proposed to compute the similarity between two graphs. Sarlin *et al.* [59] regarded the graph matching problem as an optimal transport problem and attempted to represent each graph by a graph neural network and perform graph matching using a sinkhorn algorithm.

Our method differs from these existing graph matching methods in two aspects.

1) We focus on addressing the FSRSSC problem instead of the matching problem of graph structure data. To our best knowledge, there is no previous work to explore the graph matching mechanism for FSRSSC. We fill the gap by regarding each remote sensing scene image/class as a scene graph and computing their similarity scores in a graph-matching-based manner.

2) Our method models the scene graph edges in an implicit feature encoding manner instead of the simple adjacency matrix manner, which is more helpful to model various types of spatial patters, such as the spatial pattern of "running/parked" for objects "cars" and "road" and the one of "side" for objects "grass" and "building."

III. Methodology

This article focuses on addressing the FSRSSC problem, i.e., recognizing novel scene classes with few labeled samples by leveraging base scene classes with abundant labeled samples. In this section, we first formalize the FSRSSC problem, and then introduce the proposed SGMNet framework and its core component, i.e., the scene graph matching network.

A. Problem Definition

In the FSRSSC problem, three datasets are given: a base scene dataset $\mathcal{D}_{\text{base}}$ with abundant labeled samples, a novel scene training dataset $\mathcal{S}$ (called support set) with few labeled samples, and a novel scene test dataset $\mathcal{Q}$ (called query set) consisting of unlabeled samples. In the base scene dataset $\mathcal{D}_{\text{base}}$, there is a large number of remote sensing scene images which are labeled with a base scene class $y_i \in \mathcal{C}_{\text{base}}$ where $\mathcal{C}_{\text{base}}$ is the set of base scene classes. In the support set $\mathcal{S}$, there is N novel scene classes and each scene class only contains K labeled samples for novel scene classes' learning. We denote the set of novel scene classes as $\mathcal{C}_{\text{novel}}$. Note that the base scene class set and novel scene class set are disjoint, *i.e.*, $\mathcal{C}_{\text{base}} \cap \mathcal{C}_{\text{novel}} = \emptyset$. In the query set $\mathcal{Q}$, there are some unlabeled instances from the scene classes $\mathcal{C}_{\text{novel}}$.

Our goal is to learn a good classifier $f_\theta()$ with parameters θ for query set $\mathcal{Q}$ by leveraging the support set $\mathcal{S}$ and the base scene class dataset $\mathcal{D}_{\text{base}}$. The problem is called the N-way K-shot FSRSSC problem, which can be expressed as

$$\min_{\theta} \mathbb{E}_{(x,\ y)\sim\mathcal{Q}} - \log(P(y|x,\ \theta,\ \mathcal{S},\ \mathcal{D}_{\text{base}})) \tag{1}$$

where $P(y|x,\ \theta,\ \mathcal{S},\ \mathcal{D}_{\text{base}})$ denotes the probability of classifying each test remote sensing scene image $x \in \mathcal{Q}$ to the novel scene class $y \in \mathcal{C}_{\text{novel}}$.

B. SGMNet Framework

The idea of our SGMNet is to learn a good similarity evaluation strategy from base scene classes and then transfer it to novel scene classes for performing scene class prediction in a nearest-class manner. The intuition behind such design is that similarity evaluation knowledge is class-agnostic and transferable since it only focuses on recognizing the similarity and difference between remote sensing scenes. To this end, we propose to regard each remote sensing scene image or each scene class as a scene graph, and then evaluate the similarity of each test remote sensing image and each scene class in a scene-graph-matching-based manner. The advantage of such design is that these cooccurrence objects and their spatial correlations can be explicitly captured via the scene graph nodes and their edges, thereby producing more accurate similarity evaluation.

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As shown in Fig. 2, the SGMNet framework consists of three phases: pretraining, meta-training, and meta-test phases. Next, we introduce them in detail, respectively.

1) Pretraining: We first construct a fully CNN, which consists of a feature extractor $f_{\theta_f}()$ and a softmax classification head $f_{\theta_c}()$. Then, we train the above network by minimizing the negative loglikelihood estimation on the entire base scene datasets $\mathcal{D}_{\text{base}}$ with the traditional batch-based training manner like [6]. That is,

$$\min_{\{\theta_f,\theta_c\}} \mathbb{E}_{(x,\ y)\sim\mathcal{D}_{\text{base}}} - \log(P(y|x,\ \theta_f,\ \theta_c)) \tag{2}$$

where θ_f and θ_c denote the parameters of the feature extractor $f_{\theta_f}()$ and the softmax-based classification head $f_{\theta_c}()$, respectively. Following [60], we also add a self-supervised loss, i.e., the rotation (0°, 90°, 180°, and 270°) classification loss, aiming to learn a rotation-invariant representation for each remote sensing image. Finally, we remove the softmax-based classification head $f_{\theta_c}()$ and the last global pooling layer of feature extractor $f_{\theta_f}()$, aiming to obtain a good spatial representation for each image. Note that the feature extractor $f_{\theta_f}()$ is frozen in the later meta-training and meta-test phases.

2) Meta-Training: In this phase, we design a scene graph matching network $f_{\theta_m}()$ with parameter θ_m as a meta-learner, targeting at representing each remote sensing image or each scene class as a scene graph and then evaluating their similarity in a scene graph matching manner. Refer to Section III-C for details of meta-learner. Here, we mainly focus on introducing the workflow of the meta-training phase, i.e., how to train the meta-learner $f_{\theta_m}()$ to perform few-shot scene classification tasks in a scene-graph-matching-based manner.

As shown in Fig. 2, we mimic the N-way K-shot task setting and construct a large set of few-shot remote sensing classification tasks (called episodes) from the base scene dataset $\mathcal{D}_{\text{base}}$. Specifically, for each episode, we randomly select N scene classes from the set of base scene classes $\mathcal{C}_{\text{base}}$, sample K remote sensing images per scene class from the base scene dataset $\mathcal{D}_{\text{base}}$ as support set $\mathcal{S}$, and sample M remote sensing images per scene class as query set $\mathcal{Q}$. Then, we train the meta-learner $f_{\theta_m}()$ in an episodic training manner [61].

Specifically, for each few-shot remote sensing classification task, we first leverage the pretrained feature extractor $f_{\theta_f}()$ to obtain a good spatial representation $f_{\theta_f}(x_i)$ for each remote sensing image $x_i \in \mathcal{S} \cup \mathcal{Q}$. Second, for each scene class k, we estimate its spatial representation p_k in a spatial representation average manner. That is,

$$p_k = \frac{1}{|\mathcal{S}_k|} \sum_{x_i \in \mathcal{S}_k} f_{\theta_f}(x_i) \tag{3}$$

where $\mathcal{S}_k$ denotes the remote sensing image set extracted from scene class k, and $|\cdot|$ denotes the size of a set. Third, we feed the spatial representation $f_{\theta_f}(x_i)$ (or p_k) of each query sample $x_i \in \mathcal{Q}$ (or each scene class k) into a GCM $f_{\theta_{\text{sc}}}()$, aiming to encode the spatial representation as a scene graph g_{x_i} (or g_k). That is,

$$\begin{aligned} g_{x_i} &= f_{\theta_{\text{sc}}}(f_{\theta_f}(x_i)) \\ g_{p_k} &= f_{\theta_{\text{sc}}}(p_i) \end{aligned} \tag{4}$$

where θ_{sc} denotes the parameters of the GCM $f_{\theta_{\text{sc}}}()$. Fourth, we evaluate the similarity score $s_{i,k}$ between each query sample $x_i \in \mathcal{Q}$ and each scene class k by leveraging a GMM $f_{\theta_{\text{sm}}}()$. That is,

$$s_{x_i,p_k} = f_{\theta_{\text{sm}}}(g_{x_i}, g_{p_k}), \quad s_{x_i,p_k} \in [0,1] \tag{5}$$

where θ_{sm} denotes the parameters of GMM $f_{\theta_{\text{sm}}}()$.

Finally, following [12], we take the mean-square error (MSE) as the loss function to train our meta-optimizer:

$$\min_{\theta_m} L_{\text{MSE}} = \text{MSE}(s_{x_i,p_k}, y_{x_i,k}) \tag{6}$$

where y_i is a one-hot vector converted by the label of sample x_i, and θ_m consists of θ_{sc} and θ_{sm}, i.e., $\theta_m = \{\theta_{\text{sc}}, \theta_{\text{sm}}\}$.

3) Meta-Test: In this phase, we focus on how to recognize the novel scene classes with few labeled samples in a scene-graph-matching-based manner. The workflow is similar to the meta-training phase. The only difference is that we remove the training step described in 6. We perform novel scene class prediction by evaluating the similarity between each test sample and all the scene classes using refeq4 and 5 and then assigning it to the label of the most similar scene class.

C. Scene Graph Matching Network

The key challenge of SGMNet is how to fully leverage the two characteristics of remote sensing images, i.e., the object cooccurrence and object spatial correlation, to evaluate the similarity between each test remote sensing image and each scene class. Our idea is designing a scene graph matching network to encode each remote sensing image/class as a scene graph, where the nodes reflect these cooccurrence objects; meanwhile, the edges capture the spatial correlations between these cooccurrence objects and then perform their similarity evaluation in a scene-graph-matching-based manner.

As shown in Fig. 3, the scene graph matching network consists of two key components, i.e., a GCM and a GMM. Specifically, the GCM takes the spatial representations as inputs, aiming to encode each test remote sensing image or each scene class as a scene graph. After that, the GMM takes the pair of scene graphs as inputs, accounting for using the object cooccurrence and object spatial correlation to evaluate their similarity. The advantage of such a design is that more refined information contained in cooccurrence objects and object spatial structure can be explicitly captured by the propagation/learning of layer-by-layer graph nodes and edges. As a result, a more accurate similarity evaluation can be obtained for nearest-neighbor-based scene class prediction. Next, we introduce them in detail, respectively.

1) Graph Construction Module (GCM): In the GCM, the key challenge is how to effectively encode these cooccurrence objects and their spatial relationships since it is unknown what cooccurrence objects exist in the remote sensing image or scene class. Inspired by recent works [62] in the study of interpretable convolutional networks, which reveal that the features extracted by deep convolution layer in CNN effectively capture high-level semantic information (i.e., object-level information), we directly treat the spatial representation $f_{\theta_f}(x_i)$ or p_k obtained in the last convolution layer as inputs

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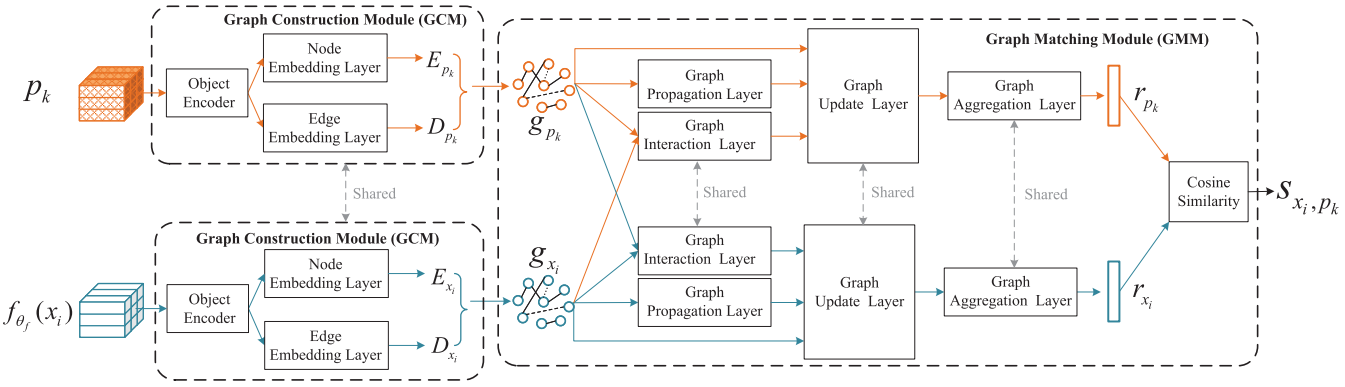


Fig. 3. Illustration of scene graph matching network, including GCM and GMM. The GCM aims to represent each image as a scene graph. The GMM accounts for computing the similarity score of each remote sensing image and each scene class. Note that the parameters of GCM and GMM are shared.

and attempt to construct a scene graph by encoding its local features as nodes and its relationships between all the local features as edges. Next, for clarity, we take the spatial representation $f_{\theta_f}(x_i)$ of remote sensing image $x_i \in \mathcal{Q}$ as an example to introduce how to construct the scene graph. The overall construction process for the scene class k is similar.

As shown in Fig. 4, given a spatial representation $f_{\theta_f}(x_i)$ of each test sample $x_i \in \mathcal{Q}$, we first encode it to a new spatial representation by an object encoder $f_{\theta_{\text{oe}}}()$. The new spatial representation will be a $W \times H \times C$ tensor, which can be considered as a set of M ($M = WH$) C-dimensional local features as $\{l_{x_i,m}\}_{m=0}^{M-1}$. Then, we use a node embedding layer $f_{\theta_{\text{ne}}}()$ to encode these local features $\{l_{x_i,m}\}_{m=0}^{M-1}$ as the node embeddings $E_{x_i} = \{e_{x_i,m}\}_{m=0}^{M-1}$, which would denote potential objects. Besides, an edge embedding layer $f_{\theta_{\text{ee}}}()$ is leveraged to explore its pairwise relationship between two local features $l_{x_i,m}$ and $l_{x_i,n}$ as the edge embeddings $D_{x_i} = \{d_{x_i,m,n}\}_{m,n=0}^{M-1}$ where $m = 0, 1, \ldots, M-1$ and $n = 0, 1, \ldots, M-1$. Finally, the above encoding process is formally expressed as

$$\begin{aligned} \{l_{x_i,m}\}_{m=0}^{M-1} &= f_{\theta_{\text{oe}}}(f_{\theta_f}(x_i)) \\ \{e_{x_i,m}\}_{m=0}^{M-1} &= f_{\theta_{\text{ne}}}\left(\{l_{x_i,m}\}_{m=0}^{M-1}\right) \\ \{d_{x_i,m,n}\}_{m,n=0}^{M-1} &= f_{\theta_{\text{ee}}}\left(\left\{l_{x_i,m}||\hat{m}||l_{x_i,n}||\hat{n}\right\}_{m=0,n=0}^{M-1}\right) \end{aligned} \tag{7}$$

where $\cdot||\cdot$ denotes a concatenation operator of two vectors, and $\hat{m}$ and $\hat{n}$ denote the one-hot vector converted by the position numbers m and n, respectively. In particular, the goal of introducing $\hat{m}$ and $\hat{n}$ as the inputs of the edge embedding layer $f_{\theta_{\text{ee}}}()$ is to model the spatial structure pattern.

As a result, we obtain a new scene-graph-based representation for each test sample $x_i \in \mathcal{Q}$, i.e., $g_{x_i} = \langle E_{x_i}, D_{x_i}\rangle$. Similarly, for each scene class k, its scene-graph-based representation $g_{p_k} = \langle E_{p_k}, D_{p_k}\rangle$ can also be obtained given its spatial representation p_k, where $E_{p_k} = \{e_{p_k,m}\}_{m=0}^{M-1}$ and $D_{p_k} = \{d_{p_k,m,n}\}_{m,n=0}^{M-1}$.

2) Graph Matching Module (GMM): Inspired by recent studies [58], [59], [63], which evaluate the similarity of graph-structured objects from the perspective of graph matching, we propose a GMM to compute the similarity score between the pair of scene graphs (i.e., g_{x_i} and g_k). To our best knowledge, there is no work to explore the graph matching mechanism for FSRSSC. In this article, we fill the gap by

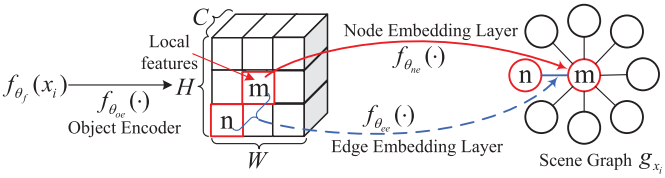


Fig. 4. Illustration of GCM. For each spatial representation $f_{\theta_f}(x_i)$, we first use an object encoder to capture cooccurrence objects, and then represent its local features as nodes and its relationships between all the local features as edges. Here, we only take nodes m and n as an example to show GCM.

encoding each remote sensing image or each scene class as a scene graph and then evaluating their similarity in a scene-graph-matching-based manner. Specifically, as shown in Fig. 3, the GMM consists of a graph propagation layer $f_{\theta_{\text{gp}}}()$ with parameter θ_{gp}, a graph interaction layer $f_{\theta_{\text{gi}}}()$ with parameter θ_{gi}, a graph update layer $f_{\theta_{\text{gu}}}()$ with parameter θ_{gu}, and a graph aggregation layer $f_{\theta_{\text{ga}}}()$ with parameter θ_{ga}. The idea of such a design is to learn a more robust representation for each test remote sensing image or each scene class by fully using its intragraph and cross-graph node and edge information, and then evaluate their similarity via a cosine similarity function. Next, we elaborate on them.

a) Graph propagation layer: To fully exploit the intragraph information (i.e., the cooccurrence objects and their spatial correlation) of each scene graph $g_{x_i} = \langle E_{x_i}, \mathcal{D}_{x_i}\rangle$, we compute a intragraph node representation $e_{x_i,m}^{\text{intra}}$ for each node m by propagating all its neighborhood node embeddings along the edge embeddings $\mathcal{D}_{x_i}$ of the scene graph g_{x_i}. Specifically, for each node $m \in [0, M-1]$, we first concatenate its node embedding $e_{x_i,m}$, the embedding $e_{x_i,n}$ of each neighborhood node $n \in [0, M-1]$, and their edge embedding $d_{x_i,m,n}$, followed by a graph propagation layer $f_{\theta_{\text{gp}}}()$ that propagates the node information on the scene graph g_{x_i}. Then, we aggregate all the neighborhood information as the intragraph node representation $e_{x_i,m}^{\text{intra}}$ in an average manner. That is,

$$e_{x_i,m}^{\text{intra}} = \frac{1}{M} \sum_{n\in[0,M-1]} f_{\theta_{\text{gp}}}(e_{x_i,m}||e_{x_i,n}||d_{x_i,m,n}) \tag{8}$$

where $\cdot||\cdot$ denotes a concatenation operator. Similarly, we can also obtain a new intragraph node representation $e_{p_k,m}^{\text{intra}}$ for the scene graph $g_{p_k} = \langle E_{p_k}, \mathcal{D}_{p_k}\rangle$ of each class k.

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b) Graph interaction layer: To fully use the cross-graph information, i.e., measuring how well a node in one scene graph (e.g., g_{x_i}) can be matched to one or more nodes in the other scene graph (e.g., g_{p_k}), we calculate a cross-graph node representation $e^{\text{cross}}_{x_i,m}$ for each node m by introducing a graph interaction layer. Specifically, we first regard these two scene graphs g_{x_i} and g_{p_k} as inputs of the graph interaction layer. Then, we compute the similarity score of cross-graph nodes by calculating the inner product of their node embeddings $e_{x_i,m}$ and $e_{p_k,n}$ where $m, n \in [0, M-1]$. Finally, a new cross-graph node representation $e^{\text{cross}}_{x_i,m}$ and $e^{\text{cross}}_{p_k,n}$ can be obtained by regarding the similarity score as matching weights, respectively. That is,

$$e^{\text{cross}}_{x_i,m} = \sum_n e_{p_k,n} \frac{\exp(e_{x_i,m} \cdot e_{p_k,n})}{\sum_n \exp(e_{x_i,m} \cdot e_{p_k,n})}$$
$$e^{\text{cross}}_{p_k,n} = \sum_m e_{x_i,m} \frac{\exp(e_{x_i,m} \cdot e_{p_k,n})}{\sum_m \exp(e_{x_i,m} \cdot e_{p_k,n})}. \tag{9}$$

c) Graph update layer: Till now, we have obtained three types of node representations for the scene graph g_{x_i} or g_{p_k}, i.e., $e_{x_i,m}$, $e^{\text{intra}}_{x_i,m}$, and $e^{\text{cross}}_{x_i,m}$, or $e_{p_k,n}$, $e^{\text{intra}}_{p_k,n}$, and $e^{\text{cross}}_{p_k,n}$. Then, we introduce a graph update layer $f_{\theta_{\text{gu}}}()$ to combine these node representations for the scene graph g_{x_i} or g_{p_k}. That is,

$$e^{\text{update}}_{x_i,m} = f_{\theta_{\text{gu}}}\left(e_{x_i,m} \middle|\middle| e^{\text{intra}}_{x_i,m} \middle|\middle| e^{\text{cross}}_{x_i,m}\right)$$
$$e^{\text{update}}_{p_k,n} = f_{\theta_{\text{gu}}}\left(e_{p_k,n} \middle|\middle| e^{\text{intra}}_{p_k,n} \middle|\middle| e^{\text{cross}}_{p_k,n}\right) \tag{10}$$

where $e^{\text{update}}_{x_i,m}$ and $e^{\text{update}}_{p_k,n}$ denote the combined node representation of the scene graphs g_{x_i} and g_{p_k}, respectively.

d) Graph aggregation layer: To evaluate the graph-level matching score, we encode the entire scene graph g_{x_i} or g_k as a graph-level representation r_{x_i} or r_k by aggregating their node-level representations. That is,

$$r_{x_i} = \sum_{m=0}^{M-1} f_{\theta_{\text{ga}}}\left(e^{\text{update}}_{x_i,m}, \frac{1}{M}\sum_{m=0}^{M-1} e^{\text{update}}_{x_i,m}\right) e^{\text{update}}_{x_i,m}$$
$$r_{p_k} = \sum_{n=0}^{M-1} f_{\theta_{\text{ga}}}\left(e^{\text{update}}_{p_k,n}, \frac{1}{M}\sum_{n=0}^{M-1} e^{\text{update}}_{p_k,n}\right) e^{\text{update}}_{p_k,n} \tag{11}$$

where $f_{\theta_{\text{ga}}}()$ denotes the graph aggregation layer, which is used for estimating the weights of node aggregation.

e) Similarity evaluation: Finally, we evaluate the similarity score between g_{x_i} and g_{p_k} by calculating the cosine similarity of two graph-level representation r_{x_i} and r_{p_k}. That is,

$$s_{x_i,p_k} = \frac{\text{cosine_similarity}(r_{x_i}, r_{p_k}) + 1}{2} \tag{12}$$

where cosine_similarity$(\cdot)$ denotes the function of cosine similarity between two vectors, and s_{x_i,p_k} is scaled to $[0, 1]$.

IV. Experiments

In this section, we evaluate the performance of the proposed SGMNet framework on four real-world datasets, and then discuss the experimental results and present our ablation study.

TABLE I

Basic Statistics of the UCMercedLandUse, WHU-RS19, NWPU-RESISC45, and AID Datasets

Datasets	The number of scene classes		
	meta-training	meta-validation	meta-test
UCMercedLandUse	10	5	6
WHU-RS19	9	5	5
NWPU-RESISC45	25	10	10
AID	16	7	7

A. Datasets and Settings

We conduct the experiments of five-way one-shot/five-shot tasks on four real-world datasets, i.e., UCMercedLandUse, NWPU-RESISC45, WHU-RS19, and AID. The details are shown in Table I. Next, we elaborate on them, respectively.

1) UCMercedLandUse: The dataset is a classical and common dataset for evaluating the performance of RSSC methods. It is composed of 21 scene classes: agricultural, airplane, baseball diamond, beach, buildings, chaparral, dense residential, forest, freeway, golf course, harbor, intersection, medium density residential, mobile home park, overpass, parking lot, river, runway, sparse residential, storage tanks, and tennis courts. For each scene class, there are 100 remote sensing images with a spatial resolution of 256 × 256. Following [21], we split the dataset into ten scene classes for meta-training, five scene classes for meta-validation, and six scene classes for meta-test, respectively.

2) WHU-RS19: The dataset is proposed by Wuhan University for RSSC, which involves 1005 remote sensing images with a spatial resolution of 600 × 600. It includes 19 scene classes, including airport, bridge, desert, football field, industrial, mountain, parking, port, residential, beach, farmland, forest, park, railway station, commercial, meadow, pond, river, and viaduct. Following [21], we split the dataset into nine scene classes for meta-training, five scene classes for meta-validation, and five scene classes for meta-test, respectively.

3) AID: The dataset is another large-scale dataset for RSSC including 30 scene classes: agricultural, airplane, baseball diamond, beach, buildings, chaparral, dense residential, forest, freeway, golf course, harbor, intersection, medium density residential, mobile home park, overpass, parking lot, river, runway, sparse residential, storage tanks, and tennis courts. In the dataset, each scene class contains around 220–420 images with the size of 600 × 600. Referring to [21], we split the dataset into 16 scene classes for meta-training, seven scene classes for meta-validation, and seven scene classes for meta-test, respectively (see Appendix for more details).

4) NWPU-RESISC45: The dataset is a large-scale RSSC dataset, which consists of 45 scene classes: airplane, airport, baseball diamond, basketball court, beach, bridge, chaparral, church, circular farmland, cloud, commercial area, dense residential, desert, forest, freeway, golf course, ground track field, harbor, industrial area, intersection, island, lake, meadow, medium residential, mobile home park, mountain, overpass, palace, parking lot, railway, railway station, rectangular farmland, river, roundabout, runway, sea ice, ship, snowberg, sparse residential, stadium, storage tank, tennis court, terrace,

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TABLE II

EXPERIMENTAL RESULTS ON THE UCMERCEDLANDUSE AND WHU-RS19 DATASETS. TOP TWO RESULTS ARE SHOWN IN BOLD AND UNDERLINE

Method	Type	Backbone	UCMerced LandUse		WHU-RS19	
			5-way 1-shot	5-way 5-shot	5-way 1-shot	5-way 5-shot
MAML [14]	Optimization	Conv-5	48.86 ± 0.74%	60.78 ± 0.62%	49.13 ± 0.65%	62.49 ± 0.51%
MetaSGD [46]	Optimization	Conv-5	50.52 ± 2.61%	60.82 ± 2.00%	51.54 ± 2.31%	61.74 ± 2.02%
LLSR [24]	Optimization	Conv-5	39.47%	57.40%	57.10%	70.65%
ProtoNet [16]	Metric	Conv-5	52.27 ± 0.20%	69.86 ± 0.15%	58.01 ± 0.16%	80.70 ± 0.11%
MatchingNet [61]	Metric	Conv-5	34.70%	52.71%	50.13%	54.10%
RelationNet [12]	Metric	Conv-5	48.08 ± 1.67%	61.88 ± 0.50%	60.92 ± 1.86%	79.75 ± 1.19%
DeepEMD [42]	Metric	Conv-5	58.47 ± 0.76%	70.42 ± 0.58%	63.76 ± 0.74%	78.25 ± 0.43%
FCASIM [64]	Metric	Conv-5	56.63 ± 0.70%	71.30 ± 0.52%	74.40 ± 0.69%	87.10 ± 0.43%
DLA-MatchNet [21]	Metric	Conv-5	53.76 ± 0.62%	63.01 ± 0.51%	68.27 ± 1.83%	79.89 ± 0.33%
RS-MetaNet [7]	Metric	Conv-5	49.68 ± 0.71%	67.53 ± 0.59%	74.58 ± 0.68%	87.45 ± 0.40%
Our Method (SGMNet)	Metric	Conv-5	**60.52 ± 0.74%**	**73.42 ± 0.49%**	**85.06 ± 0.55%**	**90.12 ± 0.27%**

TABLE III

EXPERIMENTAL RESULTS ON THE AID AND NWPU-RESISC45 DATASETS. TOP TWO RESULTS ARE SHOWN IN BOLD AND UNDERLINE

Method	Type	Backbone	AID		NWPU-RESISC45	
			5-way 1-shot	5-way 5-shot	5-way 1-shot	5-way 5-shot
MAML [14]	Optimization	Conv-5	43.20 ± 0.77%	60.37 ± 0.75%	48.40 ± 0.82%	62.90 ± 0.69%
MetaSGD [46]	Optimization	Conv-5	45.01 ± 0.98%	62.58 ± 0.80%	60.63 ± 0.90%	75.75 ± 0.65%
LLSR [24]	Optimization	Conv-5	45.18%	61.76%	51.43%	72.90%
ProtoNet [16]	Metric	Conv-5	54.32 ± 0.86%	67.80 ± 0.64%	40.33 ± 0.18%	63.82 ± 0.56%
MatchingNet [61]	Metric	Conv-5	33.87%	50.40%	37.61%	47.10%
RelationNet [12]	Metric	Conv-5	54.62 ± 0.80%	68.80 ± 0.66%	66.43 ± 0.73%	78.35 ± 0.51%
DeepEMD [42]	Metric	Conv-5	61.04 ± 0.77%	74.51 ± 0.55%	64.39 ± 0.84%	78.01 ± 0.56%
FCASIM [64]	Metric	Conv-5	61.37 ± 0.95%	74.47 ± 0.65%	67.58 ± 0.85%	81.56 ± 0.56%
DLA-MatchNet [21]	Metric	Conv-5	61.99 ± 0.94%	75.03 ± 0.67%	68.80 ± 0.70%	81.63 ± 0.46%
RS-MetaNet [7]	Metric	Conv-5	58.51 ± 0.84%	73.76 ± 0.69%	64.07 ± 0.90%	79.62 ± 0.65%
Our Method (SGMNet)	Metric	Conv-5	**62.21 ± 0.81%**	**75.68 ± 0.56%**	**70.40 ± 0.83%**	**82.32 ± 0.53%**

thermal power station, and wetland. For each scene class, there are 700 remote sensing images with a spatial resolution of 256 × 256. Following [21], we split the 45 scene classes into 25 scene classes for meta-training, ten scene classes for meta-validation, and ten scene classes for meta-test, respectively.

B. Implementation Details

1) Network Architecture: Following [21], we conduct the experiments using a shallow backbone (i.e., Conv-5) as the feature extractor, which introduces five convolution blocks for representing each remote sensing image. Here, each convolution block consists of a 3 × 3 convolution layer, a batch normalization layer, and a ReLU-based activation function layer, respectively. In particular, the first four convolution blocks additionally introduce a 2 × 2 max-pooling layers, while the last one does not. In the scene graph matching network: 1) we use a convolution block with 256 channels as the object encoder, which produces a new spatial representation with the size of 4 × 4 × 256 (UCMercedLandUse and NWPU-RESISC45) or 8 × 8 × 256 (WHU-RS19 and AID); 2) we use a two-layer fully connected network with 256 hidden units and 128-D outputs as the node embedding layer; 3) we use a two-layer fully connected network with 256 hidden units and 64-D outputs, as the edge embedding layer; 4) we use a two-layer fully connected network with 512 hidden units and 512-D outputs as the graph propagation layer; 5) we use a single-layer fully connected network with 256-D outputs as the graph update layer; and 6) we use a single-layer fully connected network as the graph aggregation layer. In all the above modules, ReLU is used as the activation function.

2) Training Details: In the pretraining phase, we use the traditional batch-based training strategy like [6] to pretrain the feature extractor with 100 epochs on the entire base scene classes using all the samples contained in the base scene class dataset (i.e., $\mathcal{D}_{\text{base}}$) and using an SGD optimizer with a momentum of 0.9 and a weight decay of 0.0005. Here, the learning rate is initialized as 0.1 and then decayed by 0.1 at epochs 60, 80, and 90, respectively. In the meta-training phase, we construct 8000 episodes by mimicking the N-way K-shot setting (i.e., five-way one-shot setting or five-way five-shot setting) to sample a large number of few-shot tasks from base scene class dataset $\mathcal{D}_{\text{base}}$ and then train our meta-learner (i.e., the scene graph matching network) with 100 epochs using an Adam optimizer with a weight decay of 0.0005 in an episodic training manner. Here, the learning rate is set to 0.00001.

3) Evaluation: We evaluate the proposed method on 600 randomly sampled five-way one-shot and five-way five-shot classification tasks from the test set and report the mean accuracy together with the 95% confidence interval. It is worth noting that we randomly sample 15 remote sensing images per class as the query set to evaluate the classification performance.

4) Baseline Methods: We select some FSNIC methods (i.e., MAML [14], MetaSGD [46], ProtoNet [16], MatchingNet [61], RelationNet [12]), and DeepEMD [42], and four state-of-the-art FSRSSC methods (i.e., LLSR [24], FCASIM [64], RS-MetaNet [7], and DLA-MatchNet [21]) as our baselines. Among these methods, MAML, MetaSGD, and LLSR can be regarded as the optimization-based meta-learning methods, and ProtoNet, MatchingNet, RN, FCASIM,

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TABLE IV
EFFECT OF BACKBONE ON THE UCMERCEDLANDUSE AND WHU-RS19 DATASETS

Method	Backbone	UCMerced LandUse		WHU-RS19	
		5-way 1-shot	5-way 5-shot	5-way 1-shot	5-way 5-shot
Our Method (SGMNet)	Conv-5 (shallow)	60.52 ± 0.74%	73.42 ± 0.49%	85.06 ± 0.55%	90.12 ± 0.27%
Our Method (SGMNet)	ResNet-12 (deep)	**64.17 ± 0.75%**	**76.63 ± 0.59%**	86.32 ± 0.54%	91.02 ± 0.30%
Our Method (SGMNet)	ResNet-34 (more deep)	58.96 ± 0.77%	68.12 ± 0.70%	**87.81 ± 0.45%**	**92.45 ± 0.22%**

TABLE V
EFFECT OF BACKBONE ON THE AID AND NWPU-RESISC45 DATASETS

Method	Backbone	AID		NWPU-RESISC45	
		5-way 1-shot	5-way 5-shot	5-way 1-shot	5-way 5-shot
Our Method (SGMNet)	Conv-5 (shallow)	62.21 ± 0.81%	75.68 ± 0.56%	70.40 ± 0.83%	82.32 ± 0.53%
Our Method (SGMNet)	ResNet-12 (deep)	**64.32 ± 0.79%**	**77.98 ± 0.42%**	**73.01 ± 0.77%**	**84.52 ± 0.50%**
Our Method (SGMNet)	ResNet-34 (more deep)	59.64 ± 0.77%	71.68 ± 0.57%	69.08 ± 0.87%	80.19 ± 0.57%

TABLE VI
ABLATION STUDY OF THE SCENE GRAPH MATCHING NETWORK ON THE UCMERCEDLANDUSE AND WHU-RS19 DATASETS

	Setting	UCMerced LandUse		WHU-RS19	
		5-way 1-shot	5-way 5-shot	5-way 1-shot	5-way 5-shot
(i)	+ Euclidean distance	54.51 ± 0.72%	71.43 ± 0.49%	75.82 ± 0.62%	84.60 ± 0.30%
(ii)	+ Cosine distance	57.38 ± 0.74%	71.96 ± 0.48%	75.52 ± 0.59%	87.64 ± 0.32%
(iii)	+ RelationNet	52.90 ± 0.73%	68.36 ± 0.53%	79.95 ± 0.44%	88.07 ± 0.38%
(iv)	+ SGMNet	**60.52 ± 0.74%**	**73.42 ± 0.49%**	**85.06 ± 0.55%**	**90.12 ± 0.27%**

TABLE VII
ABLATION STUDY OF THE SCENE GRAPH MATCHING NETWORK ON THE AID AND NWPU-RESISC45 DATASETS

	Setting	AID		NWPU-RESISC45	
		5-way 1-shot	5-way 5-shot	5-way 1-shot	5-way 5-shot
(i)	+ Euclidean distance	54.58 ± 0.83%	71.58 ± 0.61%	63.20 ± 0.89%	78.46 ± 0.56%
(ii)	+ Cosine distance	58.53 ± 0.81%	72.33 ± 0.61%	66.74 ± 0.88%	80.51 ± 0.53%
(iii)	+ RelationNet	58.75 ± 0.84%	72.10 ± 0.67%	65.58 ± 0.83%	79.20 ± 0.60%
(iv)	+ SGMNet	**62.21 ± 0.81%**	**75.68 ± 0.56%**	**70.40 ± 0.83%**	**82.32 ± 0.53%**

RS-MetaNet, and DLA-MatchNet can be regarded as the metric-based meta-learning methods.

C. Discussion of Results

In this section, we report the experimental results (five-way one-shot and five-way five-shot tasks) of all the baseline methods and the proposed SGMNet on two small-scale datasets (i.e., UCMercedLandUse and WHU-RS19) and two large-scale datasets (i.e., AID and NWPU-RESISC45). These experimental results are shown in Tables II and III.

1) Results on Small-Scale Dataset: Table II shows the experimental results of the baselines and the proposed SGMNet on the UCMercedLandUse and WHU-RS19 datasets. From Table II, we can see that the proposed SGMNet outperforms all the baseline methods, by around 1%–5% on the classification accuracy of the five-way one-shot and five-shot tasks. This verifies the effectiveness of our method. Specifically, compared with the optimization-based approaches, our method achieves 1%–27% higher classification accuracy. Different from these optimization-based methods, our method follows the framework of metric-based few-shot learning and proposes to compute the similarity score in the scene graph level, instead of learning a good initialization or optimization algorithm over few labeled samples. These experimental results show the superiority of the proposed SGMNet method. Different from the metric-based approaches, our method explores the object cooccurrence and object spatial correlation by regarding each remote sensing image or each scene class as a scene graph. The experimental results demonstrate that the proposed SGMNet is effective, with an improvement of 1%–10%. Finally, we would like to emphasize that compared with these state-of-the-art FSRSSC methods (i.e., LLSR, FCASIM, RS-MetaNet, and DLA-MatchNet), our method outperforms them by a large margin, around 10%–15%. The reasons may be that our method 1) regards each remote sensing scene image or each scene class as a scene graph and 2) evaluates their similarity scores in a scene-graph-matching-based manner, which fully leverages the object cooccurrence and object spatial correlation of remote sensing images. As we analyze in the Fig. 1, the object cooccurrence and spatial correlation are the unique advantage of remote sensing scene images.

2) Results on Large-Scale Dataset: Table III shows the experimental results of the baseline methods and the proposed SGMNet method on the AID and NWPU-RESISC45 datasets. We can also observe that our method achieves superior performance with an improvement of 1%–5% and 1%–6% in the five-way one-shot and five-way five-shot tasks, respectively. This further verifies the effectiveness of our method. Specifically; 1) our method also outperforms these optimization-based methods by around 4%–10%. This further shows the superiority of the designed scene graph matching strategy and 2) compared with the metric-based FSNIC methods (e.g., ProtoNet, MatchingNet, DeepEMD, and RelationNet) and the state-of-the-art FSRSSC methods (e.g., RS-MetaNet and DLA-MatchNet), our method also achieves 3%–9% higher accuracy. This further verifies the universality

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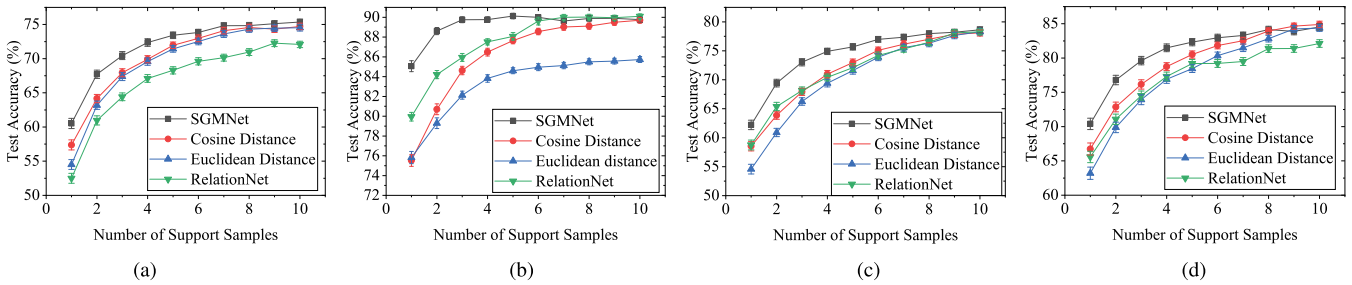


Fig. 5. Experimental results of the five-way FSRSSC task setting with different numbers of support samples on the four real-world datasets. (a) UCMercedLandUse. (b) WHU-RS19. (c) AID. (d) NWPU-RESISC45.

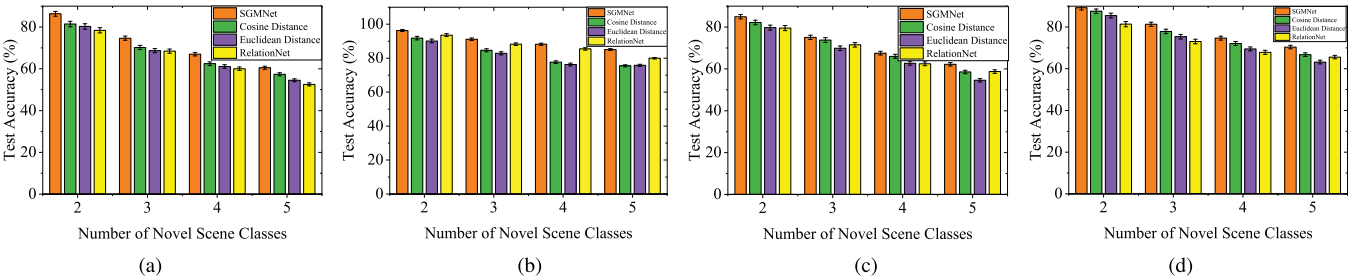


Fig. 6. Experimental results of the one-shot FSRSSC task setting with different numbers of novel scene classes on the four real-world datasets. (a) UCMercedLandUse. (b) WHU-RS19. (c) AID. (d) NWPU-RESISC45.

and effectiveness of our SGMNet in the large-scale remote sensing scene dataset, which exhibits richer remote sensing scene classes.

D. Ablation Study

In this section, we conduct the experiments of ablation study to analyze the effectiveness of different-scale backbone, scene graph matching network, two key components (i.e., graph propagation layer and graph interaction layer) of GMM, different support set sizes, and different numbers of novel classes. The experiments are summarized as answering five questions.

1) How Does the Backbone Affect Classification Performance: In Tables IV and V, we conduct an ablation study to analyze the impacts of different-scale backbone (i.e., Conv-5, ResNet-12, ResNet-34) on the small-scale datasets (UCMercedLandUse and WHU-RS19) and large-scale datasets (AID and NWPU-RESISC45), respectively. It can be found that 1) the classification performance is further improved when a deep backbone (ResNet-12) is used in our framework, around 2%–4%. This is because these deep backbones have higher performance on feature representation; but 2) when we replace the deeper ResNet34 as backbone, the performance of our SGMNet decreases by around 1%–6% except for the WHU-RS19 dataset. The experimental results imply that the ResNet34 backbone with more learnable parameters may tend to overfit the base scene class data in the pretraining phase, which results in poor generalization on novel scene classes.

2) Is the Scene Graph Matching Network Effective: In Tables VI and VII, we evaluate the classification performance of different metric methods on the pretrained feature extractor, including nonparametric metric (i.e., Euclidean distance and cosine distance) and learnable metric (i.e., RelationNet). Note that our method uses a scene graph matching network to evaluate the similarity score in the scene graph level. Thus, it can be regarded as a learnable metric method. From Tables VI and VII, we observe that: 1) compared with these nonparametric metric methods, our method achieves 1%–3% higher classification accuracy and 2) our method also beats the learnable metric method (i.e., RelationNet), with an improvement of 5%–8%. Different from these metric methods, our method effectively exploits the unique advantages of remote sensing scene images, i.e., object cooccurrence and object spatial correlation, by regarding each remote sensing scene image as a scene graph. These experimental results demonstrate the effectiveness of our method.

3) Is the Two Key Components (Graph Propagation Layer and Graph Interaction Layer) Effective in GMM: In Tables VIII and IX, we evaluate the effect of the two key components by removing them in the proposed SGMNet, respectively. Specifically, 1) we remove the graph propagation layer on GMM and revise the input dimensions of the graph update layer to 512 and 2) we remove the graph interaction layer on GMM and revise the input dimensions of the graph update layer to 512. From the results of setting 1) and 2) of Tables VIII and IX, we observe that the classification performance of the proposed SGMNet decreases by 1%–2% when removing the two components, respectively. These experimental results imply that using the graph propagation layer and the graph interaction layer is helpful for our method.

4) How Does the Number of Labeled Samples Affect Classification Performance: We conduct a statistical experiment on the UCMercedLandUse, WHU-RS19, AID, and NWPU-RESISC45 datasets by varying the number of labeled samples per class (i.e., K) from $K = 1$ to $K = 10$. The experimental results are shown in Fig. 5. From Fig. 5, we find that: 1) the classification performance of our method gradually increases with the number increase in labeled samples. This

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TABLE VIII

ABLATION STUDY OF THE SCENE GRAPH MATCHING NETWORK ON THE UCMERCEDLANDUSE AND WHU-RS19 DATASETS

	Setting	UCMerced LandUse		WHU-RS19	
		5-way 1-shot	5-way 5-shot	5-way 1-shot	5-way 5-shot
(i)	Our method (SGMNet)	**60.52 ± 0.74**%	**73.42 ± 0.49**%	**85.06 ± 0.55**%	**90.12 ± 0.27**%
(ii)	w/o graph propagation layer	59.02 ± 0.71%	72.69 ± 0.48%	84.13 ± 0.57%	89.83 ± 0.26%
(iii)	w/o graph interaction layer	58.85 ± 0.71%	71.38 ± 0.46%	83.40 ± 0.53%	88.40 ± 0.27%

TABLE IX

ABLATION STUDY OF THE SCENE GRAPH MATCHING NETWORK ON THE AID AND NWPU-RESISC45 DATASETS

	Setting	AID		NWPU-RESISC45	
		5-way 1-shot	5-way 5-shot	5-way 1-shot	5-way 5-shot
(i)	Our method (SGMNet)	**62.21 ± 0.81**%	**75.68 ± 0.56**%	**70.40 ± 0.83**%	**82.32 ± 0.53**%
(ii)	w/o graph propagation layer	61.53 ± 0.82%	74.84 ± 0.60%	67.85 ± 0.84%	81.39 ± 0.55%
(iii)	w/o graph interaction layer	61.22 ± 0.81%	73.16 ± 0.61%	67.46 ± 0.79%	81.21 ± 0.56%

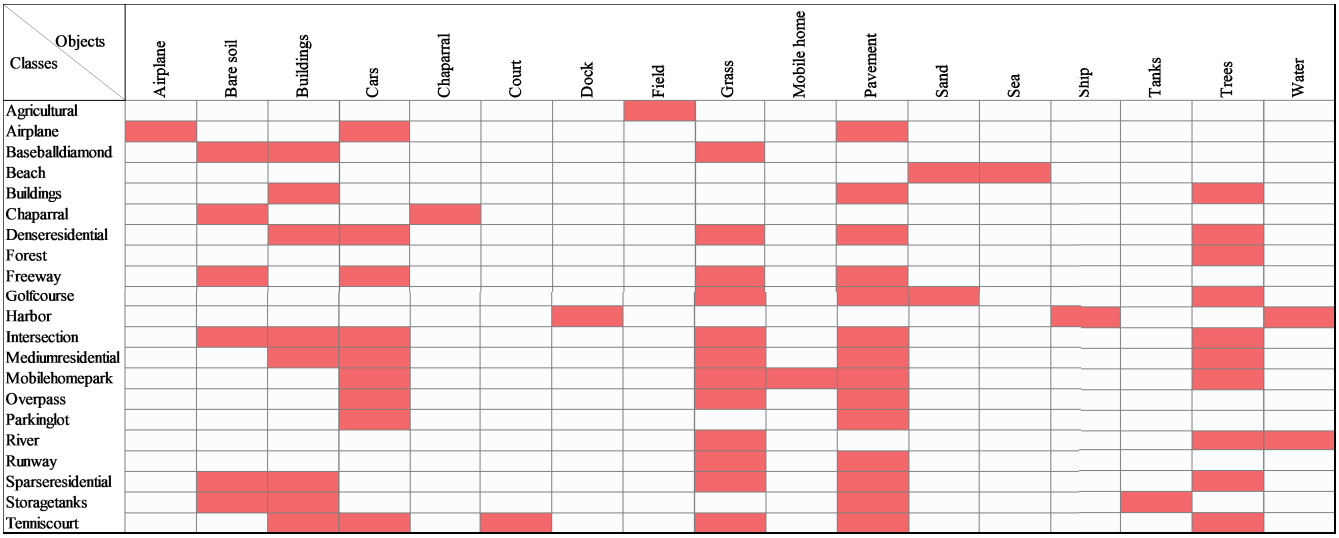


Fig. 7. Statistics of cooccurrence objects on the UCMercedLandUse dataset. The results are obtained using the object-level label annotations in [65].

is reasonable because more discriminated features can be obtained from these labeled samples; 2) our method outperforms the existing methods, especially when only few labeled samples are available (e.g., $K = 1{\sim}5$); and 3) our method performs slightly worse than the existing methods on the WHU and NWPU-RESISC45 datasets when $K = 8{\sim}10$. These results implies that exploring object cooccurrence and object spatial correction is more beneficial for FSRSSC, when very few labeled remote sensing images are available (e.g., $K = 1{\sim}8$). This is because these unique characteristics can provide more refined description for encoding each scene class, thereby effectively alleviating the data scarcity issue of FSRSSC.

5) Is Our Method Effective on Different Numbers of Novel Scene Classes: In Fig. 6, we report the classification performance of our method on the one-shot tasks by varying the number of novel scene classes from $N = 2$ to $N = 5$. It can be found that: 1) the classification performance of our method decreases as the class number N increases and 2) our method outperforms the existing method on all the few-shot task settings by around 1%–10%. This means that our method is robust and effective for various N-way classification task settings.

6) How Are These Cooccurrence Objects Distributed in Different Scene Classes: We count the cooccurrence objects on the real-world UCMercedLandUse dataset using the object-level multiple labels annotated in [65]. The statistical results are shown in Fig. 7. Here, the objects that appeared in more than 50% of the images of each remote sensing scene class are marked by a red rectangle. From the statistical results of appeared object, we can see that: 1) multiple objects tend indeed to appear together in the remote sensing image for most scene classes and 2) these cooccurrence objects are different in different remote sensing scene classes. This means that exploring cooccurrence objects is reasonable, which is helpful for FSRSSC since it provides more refined discriminable information to alleviate the data scarcity issue of FSRSSC.

7) What Is the Computational Complexity and Running Time of Our SGMNet: In Table X, we select some state-of-the-art FSRSSC methods (i.e., LLSR, DeepEMD, FCASIM, and RS-MetaNet) as comparison baselines and report the inference time and GPU memory of baseline methods and our SGMNet. All the results are averaged on 600 episodes and counted on the same machine with an NVIDIA RTX8000 GPU. From the results, we can see that 1) our method is more efficient than the optimization-based meta-learning methods

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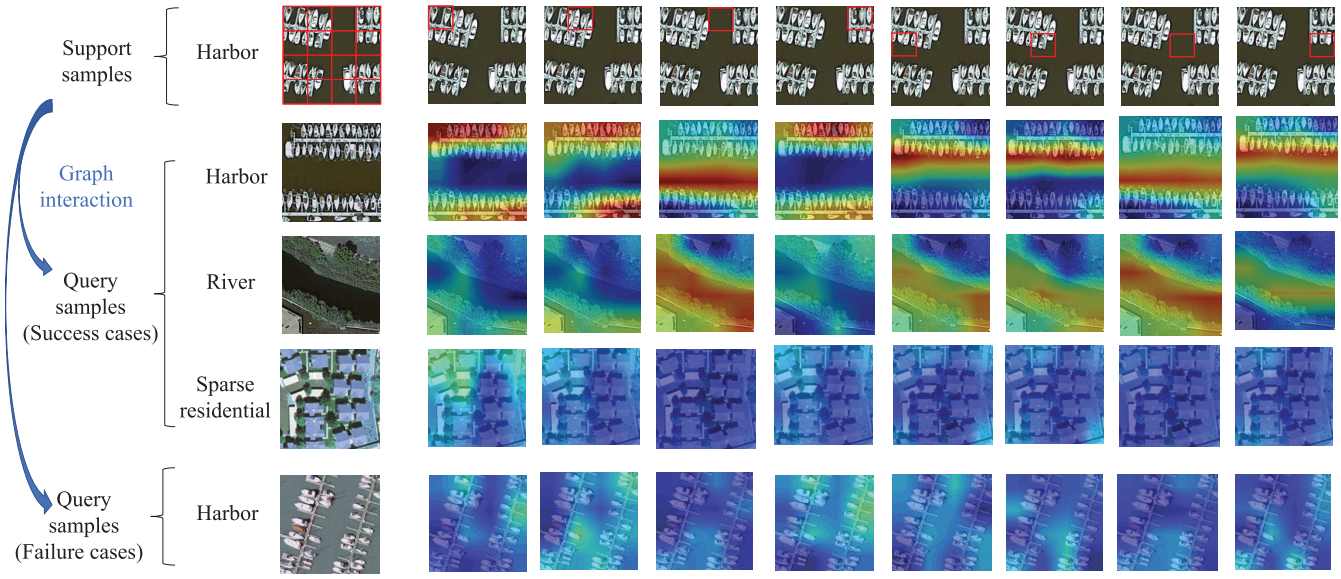


Fig. 8. Visualization of the graph matching mechanism of SGMNet. The first column represents the original images. The subsequent eight columns show the matching results of each local feature (i.e., each node of scene graph) represented by a red square box in the supporting image with the query images. Note that the matching results are shown using a bilinear interpolation manner, i.e., upsampling the matching weights of graph interaction layer to the size of query images. We can see that 1) in the success cases, the matching weights can align the same local features (i.e., potential cooccurrence objects such as ships and water) well when the support image and the query image match or partially match. Meanwhile, it tends to produce relatively low matching weights when they do not match and 2) in the failure cases, due to the large intraclass variances, the cooccurrence objects, especially water, cannot match well.

TABLE X

AVERAGED INFERENCE TIME (RUNNING TEN TIMES) OF EACH EPISODE FOR BASELINES AND OUR SGMNET ON THE UCMERCEDLANDUSE DATASET

	Method	Inference Time	GPU memory
(i)	LLSR [24]	0.1578 s	7.726 G
(ii)	DeepEMD [42]	0.2560 s	2.894 G
(iii)	FCASIM [64]	0.0826 s	2.904 G
(iv)	RS-MetaNet [7]	0.0908 s	2.970 G
(v)	Our method (SGMNet)	0.1044 s	2.906 G

(LLSR) on inference time and GPU memory and 2) takes comparable inference time and GPU memory with the metric-based meta-learning methods (DeepEMD, FCASIM, and RS-MetaNet).

E. Visualization Analysis

In this section, we show how the graph matching mechanism works by visualizing the matching weights of the graph interaction layer. The result on the test scene classes of "harbor," "river," and "sparse residential" from the UCMercedLandUse dataset is shown in Fig. 8. Here, the proposed SGMNet is trained by following the five-way one-shot task setting. Specifically, in the visualization experiment, we randomly select a remote sensing image from the scene class of "harbor" as the support sample, and three remote sensing images from the scene class of "harbor," "river," and "sparse residential," respectively, as the query samples in the success cases. Besides, a misidentified remote sensing image of scene class "harbor" is also randomly selected as the query image to show the failure case. It is worth noting that the harbor image has two cooccurrence objects, i.e., ships and water. The river image contains three cooccurrence objects (i.e., grass, trees, and water). Here, the object of "water" is overlapped with the cooccurrence objects of the harbor image. In the sparse residential image, there are five cooccurrence objects, i.e., grass, trees, buildings, cars, and pavement, and we note that none of these five objects appears in the harbor image. Then, we show the matching weights of the graph interaction layer (i.e., the details of matching between each potential object of the support sample and all the potential objects of the query samples) in Fig. 8. Note that each node of scene graph is related to the local feature of the image, which would denote a potential object.

From Fig. 8, we find that 1) for the success cases, the matching weights can align the same objects (i.e., cooccurrence objects such as ships and water) well when the two remote sensing images match or partially match, while tending to produce relatively low matching weights when they do not match. This visualization results indicate that the proposed graph matching mechanism is helpful for FSRSSC, which can effectively align and highlight the same objects appearing in two remote sensing images and provide more refined matching detail information for performing the similarity evaluation between two remote sensing scene images and 2) for the failure case, the cooccurrence objects of support sample, especially water, cannot match well with the query sample of "harbor." Such failure matching may be caused by the large distribution difference in the query sample (see the top of Fig. 8) with support sample (see the end of Fig. 8), i.e., the image difference in object "water" is very large on the support sample and the query sample due to different geographical areas, different ecological environment, or remote sensing instrument attenuation. However, we would like to emphasize that in this article, we mainly focus on learning a better similarity evaluation of remote sensing scene classes for FSRSSC. How to alleviate such distribution difference for achieving better graph matching is very interesting. We will attempt to address the issue in our future work by learning a more robust spatial representation for each remote sensing image.

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Fig. 9. Example of bad case caused by spurious correlations of cooccurrence objects on the UCMercedLandUse dataset.

F. Discussion of Spurious Correlation Issue on Our SGMNet

The idea of our SGMNet is leveraging the two unique characteristics (i.e., object cooccurrence and object spatial correlation) of remote sensing scene image to alleviate the data scarcity issue of FSRSSC since the two characteristics can provide more refined descriptions for each scene class. However, we note that two recent studies [27], [28] point out that these cooccurrence objects may introduce a spurious correlation issue between the features of cooccurrence objects and labels of support samples, which impedes few-shot learning performance. To verify whether this spurious correlation issue also exists in our SGMNet, we further conduct a bad case study experiment on the two-way five-shot tasks of the UCMercedLandUse dataset. Specifically, following [28], we select two scene classes (i.e., "golf course" and "river") and five remote sensing images with latent spurious correlation issue per class as the support set and 15 remote sensing images per class as the query set. Then, we visualize all the support images and four misclassified query images in Fig. 9 for the bad case study.

From Fig. 9, we can see that all the support images of "golf course" scene class contain cooccurrence objects of green grass and trees, but the ones of river scene class contain cooccurrence objects of yellow grass and trees, which causes spurious correlations between the cooccurrence object features and support labels such that these query images (i.e., two golf course scene images with yellow grass and trees and two river scene images with green grass and trees) are classified as wrong classes. The above analysis of the bad case study suggests that the spurious correlation issue also exists in our SGMNet, which may limit the performance of our SGMNet. However, it is worth noting that this is only a special case which may occur in a low probability, especially in some settings that more labeled samples are available. Thus, our idea actually can well work in most cases. In future work, we will attempt to address this special issue for further improving the performance of our SGMNet.

V. Conclusion

In this article, we identify two unique characteristics of remote sensing images for FSRSSC, i.e., object cooccurrence and object spatial correction, which can effectively alleviate the data scarcity issue. Based on this, we propose a novel scene-graph-matching-based meta-learning framework for FSRSSC. In particular, the remote sensing image/class is regarded as a scene graph, and then a novel scene graph matching network is carefully designed to compute the similarity between each test remote sensing image and each scene class in a scene graph matching manner. As a result, the novel scene class can be predicted through a nearest-class classifier. The experiments on four remote sensing datasets show that our model obtains significantly superior performance over the state-of-the-art methods. We also conduct extensive ablation studies and visualization analysis, which further show the superiority of our method, especially when only very few labeled samples are available.

This article mainly focuses on improving the performance of nearest-class classifier for FSRSSC by learning a good similarity evaluation between each remote sensing image and scene class in a scene graph matching manner. In future, 1) we will further explore the characteristics of object cooccurrence and object spatial correlation on other classifiers and learn a more robust spatial representation to eliminate distribution difference in remote sensing images for FSRSSC and 2) we will attempt to address the spurious correlation issue between the features of cooccurrence objects and the support labels for further improving the performance of our SGMNet.

Appendix
Details of Datasets

In the section of experiments, we evaluate some baseline methods and our SGMNet method on four real-world remote sensing datasets, i.e., UCMercedLandUse, NWPU-RESISC45, WHU-RS19, and AID. Here, for the UCMercedLandUse, NWPU-RESISC45, and WHU-RS19 datasets, we split them into three scene class sets, i.e., meta-training, meta-validation, and meta-test. The split strategy is following the split strategies proposed by [21]. Refer to their original article for more details. It is worth noting that for the AID dataset, we use a random manner to split it into 16 scene classes for meta-training, seven scene classes for meta-validation, and seven scene classes for meta-test, respectively. Specifically, the meta-training class set contains 16 scene classes, i.e., playground, stadium, resort, mountain, center, port, bridge, storage tanks, park, square, farmland, desert, commercial, railway station, beach, and parking; the meta-validation class set contains seven scene classes, i.e., church, meadow, baseball field, school, river, bare land, and pond; and the meta-test class set contains seven scene classes, i.e., viaduct, industrial, dense residential, medium residential, airport, forest, and sparse residential.

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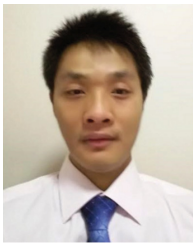
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